

Lecture 5: RBC with Endogenous Labor

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HP filter

HP filter: decompose economic series into trend and cyclical component

$$y_t = y_t^c + y_t^T$$
$$\min_{\{y_t^T\}_{t=1}^N} \mathcal{L} = \underbrace{\sum_{t=1}^N (y_t - y_t^T)^2}_{(1)} + \lambda \underbrace{\sum_{t=2}^{N-1} \left[(y_{t+1}^T - y_t^T) - (y_t^T - y_{t-1}^T) \right]^2}_{(2)}$$

Notes:

- 1) penalty of calling everything trend
- 2) change in trend growth rate, penalty of wiggles

If λ huge, close to straight line in logs.

If $\lambda = 0$, wiggle a lot, fit to y_t .

Define

$$\Delta^2 y_t^T \equiv y_{t+1}^T - 2y_t^T + y_{t-1}^T \quad (\text{second-order difference})$$

$$\frac{\partial \mathcal{L}}{\partial y_t^T} = 0 \quad \Rightarrow \quad \frac{\partial (1)}{\partial y_t^T} = 2(y_t^T - y_t)$$

Since

$$\Delta^2 y_{t-1}^T = y_t^T - 2y_{t-1}^T + y_{t-2}^T \quad (\text{coefficient } +1)$$

$$\Delta^2 y_t^T = y_{t+1}^T - 2y_t^T + y_{t-1}^T \quad (\text{coefficient } -2)$$

$$\Delta^2 y_{t+1}^T = y_{t+2}^T - 2y_{t+1}^T + y_t^T \quad (\text{coefficient } +1)$$

$$\begin{aligned} \frac{\partial}{\partial y_t^T} \sum_{s=2}^{N-1} (\Delta^2 y_s^T)^2 &= 2\Delta^2 y_{t-1}^T \frac{\partial \Delta^2 y_{t-1}^T}{\partial y_t^T} + 2\Delta^2 y_t^T \frac{\partial \Delta^2 y_t^T}{\partial y_t^T} + 2\Delta^2 y_{t+1}^T \frac{\partial \Delta^2 y_{t+1}^T}{\partial y_t^T} \\ &= 2(\Delta^2 y_{t-1}^T - 2\Delta^2 y_t^T + \Delta^2 y_{t+1}^T) \end{aligned}$$

Since

$$\begin{aligned}\Delta^2 y_{t-1}^T &= y_t^T - 2y_{t-1}^T + y_{t-2}^T, \\ -2\Delta^2 y_t^T &= -2(y_{t+1}^T - 2y_t^T + y_{t-1}^T) = -2y_{t+1}^T + 4y_t^T - 2y_{t-1}^T, \\ \Delta^2 y_{t+1}^T &= y_{t+2}^T - 2y_{t+1}^T + y_t^T,\end{aligned}$$

we get

$$\frac{\partial \mathcal{L}}{\partial y_t^T} = \frac{\partial(1)}{\partial y_t^T} + \lambda \frac{\partial}{\partial y_t^T} \sum_{s=2}^{N-1} (\Delta^2 y_s^T)^2 = 0.$$

$$\Rightarrow (1 + 6\lambda)y_t^T - 4\lambda y_{t-1}^T - 4\lambda y_{t+1}^T + \lambda y_{t-2}^T + \lambda y_{t+2}^T = y_t, \quad 3 \leq t \leq N-2.$$

At $t = 1$ (only $\Delta^2 y_2^T$ exists),

$$(1 + \lambda)y_1^T - 2\lambda y_2^T + \lambda y_3^T = y_1.$$

At $t = 2$ (only $\Delta^2 y_2^T$ and $\Delta^2 y_3^T$ exist),

$$-2\lambda y_1^T + (1 + 5\lambda)y_2^T - 4\lambda y_3^T + \lambda y_4^T = y_2.$$

Symmetrically for $t = N$ and $t = N - 1$, collect $\{y_t\}$ into the $N \times 1$ vector y and $\{y_t^T\}$ into y^T , then we have

$$Ay^T = y \quad \Rightarrow \quad y^T = A^{-1}y \quad \Rightarrow \quad y^c = y - A^{-1}y.$$

where

$$A = \begin{pmatrix} 1 + \lambda & -2\lambda & \lambda & 0 & 0 & \cdots & 0 & 0 & 0 \\ -2\lambda & 1 + 5\lambda & -4\lambda & \lambda & 0 & \cdots & 0 & 0 & 0 \\ \lambda & -4\lambda & 1 + 6\lambda & -4\lambda & \lambda & \cdots & 0 & 0 & 0 \\ 0 & \lambda & -4\lambda & 1 + 6\lambda & -4\lambda & \ddots & 0 & 0 & 0 \\ \vdots & & \ddots & \ddots & \ddots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda & -4\lambda & 1 + 6\lambda & -4\lambda & \lambda \\ 0 & 0 & 0 & \cdots & 0 & \lambda & -4\lambda & 1 + 5\lambda & -2\lambda \\ 0 & 0 & 0 & \cdots & 0 & 0 & \lambda & -2\lambda & 1 + \lambda \end{pmatrix}.$$

Let D be the $(N - 2) \times N$ second-difference matrix, whose t -th row ($t = 2, \dots, N - 1$) looks like

$$[0 \ \cdots \ 1 \ -2 \ 1 \ \cdots \ 0]$$

so that

$$(Dy^T)_{t-1} = \Delta^2 y_t^T.$$

The HP objective becomes

$$\min_{y^T} (y - y^T)'(y - y^T) + \lambda(Dy^T)'(Dy^T).$$

F.O.C.

$$-2(y - y^T) + 2\lambda D' D y^T = 0 \quad \Rightarrow \quad (I + \lambda D' D) y^T = y.$$

Hence

$$A = I + \lambda D' D.$$

As a 5-diagonal pattern:

$$\begin{array}{l} \text{interior diagonal: } [0 \ \cdots \ 1 \ -2 \ 1 \ \cdots \ 0] \\ \text{first-off diagonal: } [0 \ \cdots \ 1 \ -2 \ 1 \ 0 \ \cdots \ 0] \\ \text{second-off diagonal: } [0 \ \cdots \ 1 \ -2 \ 1 \ 0 \ 0 \ \cdots \ 0] \end{array} \begin{bmatrix} 0 \\ \vdots \\ 1 \\ -2 \\ 1 \\ \vdots \\ 0 \end{bmatrix} = 6$$

$$\begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ -2 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = -4$$

$$\begin{bmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ 1 \\ -2 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 1$$

$$\lambda \uparrow \quad \Rightarrow \quad \text{the smoother the series } y_t^T.$$

Pros

- 1) Simple and transparent. (defined by one objective)
- 2) Produces a smooth trend with tunable smoothness λ
- 3) Widely used in Macro

Cons

- 1) End-point problem: start and end of sample is distorted.
- 2) λ sensitivity: moments (volatility, correlations) change a lot with λ .
- 3) Spurious dynamics: HP can induce artificial serial correlations and distort cross-corr between series (create cycles).
- 4) Leakage: low-frequency cycles leak into trend.
- 5) Not structural: no model interpretation.

Why $\lambda = 1600$ is the quarterly standard?

Rule of thumb:

$$\lambda = 100 \times (\text{periods per year})^2.$$

Hence

$$\lambda_{\text{quarter}} = 100 \times 4^2 = 1600, \quad \lambda_{\text{month}} = 100 \times 12^2 = 14400, \quad \lambda_{\text{annual}} = 100 \times 1^2 = 100.$$

Long and Plosser (1983, JPE)

Social planner's Problem Setup:

$$U = \max_{\{C_t, L_t, K_t\}} E_0 \left\{ \sum_{t=0}^{\infty} \beta^t [\theta \ln C_t + (1 - \theta) \ln L_t] \right\}$$

s.t.

$$K_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha} - C_t \quad \text{and } K_0 \text{ is given.}$$

Resource constraint:

$$C_t + K_t = Y_t.$$

$$N_t + L_t = 1 \quad (\text{trade-off})$$

L : leisure N : working hour.

Total time is normalized to 1.

where

$$\log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t, \quad \varepsilon_t \stackrel{i.i.d.}{\sim} N(0, \sigma^2), \quad \psi < 1.$$

Note: capital completely turns over,

$$(1 - \delta)K_{t-1} = 0 \quad \delta = 1.$$

Lagrangian:

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[\theta \ln C_t + (1 - \theta) \ln(1 - N_t) + \lambda_t (Z_t K_{t-1}^\alpha N_t^{1-\alpha} - C_t - K_t) \right].$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \quad \Rightarrow \quad \frac{\theta}{C_t} = \lambda_t \quad (1)$$

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial N_t} = 0 \quad &\Rightarrow \quad -\underbrace{\frac{1 - \theta}{1 - N_t}}_{\text{utility loss}} + \underbrace{\lambda_t}_{\text{MUC}_t} \underbrace{(1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha}}_{\text{MPN}_t} = 0 \\ &\Rightarrow \quad \frac{1 - \theta}{1 - N_t} = (1 - \alpha) \lambda_t Z_t K_{t-1}^\alpha N_t^{-\alpha} \end{aligned} \quad (2)$$

labor-leisure condition (intratemporal)

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial K_t} = 0 \quad &\Rightarrow \quad -\lambda_t \beta^t + \beta^{t+1} E_t \left[\lambda_{t+1} \frac{\partial}{\partial K_t} (Z_{t+1} K_t^\alpha N_{t+1}^{1-\alpha}) \right] = 0 \\ &\Rightarrow \quad \lambda_t = \beta E_t (\lambda_{t+1} \alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha}) \end{aligned} \quad (3)$$

Euler equation (intertemporal)

TVC:

$$\lim_{T \rightarrow \infty} \beta^T U_C(C, L) K_{T+1} = 0. \quad (4)$$

Closed-form solution: constant savings rate and constant labor

Tractability: log utility + Cobb–Douglas + full depreciation

$\Rightarrow$ constant savings rate / hours.

Since

$$Y_{t+1} = Z_{t+1} K_t^\alpha N_{t+1}^{1-\alpha},$$

we have

$$\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} = \alpha \frac{Y_{t+1}}{K_t}.$$

Plug into (3):

$$\lambda_t = \beta E_t \left(\lambda_{t+1} \alpha \frac{Y_{t+1}}{K_t} \right).$$

Using (1):

$$\begin{aligned} \frac{\theta}{C_t} &= \beta E_t \left(\frac{\theta}{C_{t+1}} \alpha \frac{Y_{t+1}}{K_t} \right) \\ \Rightarrow \quad \frac{1}{C_t} &= \beta \alpha E_t \left[\frac{Y_{t+1}}{C_{t+1}} \frac{1}{K_t} \right]. \end{aligned} \tag{5}$$

Guess and verify:

$$C_t = (1 - s)Y_t, \quad s \text{ is the savings rate}, \quad K_t = sY_t. \quad (\text{full depreciate})$$

Then

$$\frac{Y_{t+1}}{C_{t+1}} = \frac{1}{1 - s}.$$

Plug into (5):

$$\begin{aligned} \frac{1}{(1 - s)Y_t} &= \alpha \beta \frac{1}{1 - s} \frac{1}{sY_t} \\ \Rightarrow \quad s &= \alpha \beta. \end{aligned}$$

Hence

$$C_t = (1 - \alpha \beta)Y_t, \quad K_t = \alpha \beta Y_t.$$

$$\begin{cases} \alpha \uparrow \Rightarrow \text{capital income share} \uparrow \Rightarrow \text{savings rate} \uparrow, \\ \beta \uparrow \Rightarrow \text{discount factor} \downarrow \Rightarrow \text{savings rate} \uparrow. \end{cases}$$

Plug into (2):

$$\begin{aligned} \frac{1 - \theta}{1 - N_t} &= (1 - \alpha) \frac{\theta}{C_t} \frac{Y_t}{N_t} = (1 - \alpha) \frac{\theta}{(1 - \alpha \beta)Y_t} \frac{Y_t}{N_t} \\ \Rightarrow \quad (1 - \theta)(1 - \alpha \beta)N_t &= (1 - \alpha)\theta(1 - N_t) \\ \Rightarrow \quad N_t = \bar{N} &= \frac{\theta(1 - \alpha)}{\theta(1 - \alpha) + (1 - \theta)(1 - \alpha \beta)}. \end{aligned}$$

Since $N_t = \text{constant } \forall t$, we have

$$\hat{n}_t = 0.$$

Also

$$K_t = \alpha\beta Y_t, \quad C_t = (1 - \alpha\beta)Y_t$$

so

$$\hat{k}_t = \hat{c}_t = \hat{y}_t.$$

Since

$$\hat{y}_t = \hat{z}_t + \alpha\hat{k}_{t-1} + (1 - \alpha)\hat{n}_t = \hat{z}_t + \alpha\hat{k}_{t-1},$$

we get

$$\hat{k}_t = \hat{c}_t = \hat{y}_t = \hat{z}_t + \alpha\hat{k}_{t-1}.$$

And

$$\hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t.$$

TFP shock alone can generate output/consumption/investment fluctuations, but

- 1) labor does not fluctuate at all,
- 2) c, k, y move one-for-one.

Consumption is usually smoother than output; capital is usually more volatile.

- 3) Labor productivity moves like output in model.

$$\left(\frac{\hat{Y}}{\hat{N}} \right)_t = \hat{y}_t - \hat{n}_t = \hat{y}_t, \quad \text{while in data,}$$

$$\sigma(Y/N) \ll \sigma(Y), \quad \text{and } \text{Corr}(Y/N, Y) = 0.55.$$

- 4) Prices (MRS, MRT, MPL, MPK) too-tight with output & TFP, hard to generate weak wage cyclical & interest rate counter-cyclical.

Labor market equilibrium

Household's problem:

$$U_{N,t} + \lambda_t W_t = 0 \quad \Rightarrow \quad U_{N,t} + U_{C,t} W_t = 0$$

labor supply

Firm's problem:

$$\max_{K_{t-1}, N_t} \Pi_t = Y_t - W_t N_t - R_t K_{t-1}$$

labor demand

Hence

$$W_t = Z_t \frac{Y_t}{N_t} (1 - \alpha).$$

From household labor supply:

$$W_t = \frac{1 - \theta}{\theta} \frac{C_t}{1 - N_t}.$$

Equivalently,

$$W_t = \frac{1 - \theta}{\theta} \frac{(1 - \alpha\beta) Z_t K_{t-1}^\alpha N_t^{1-\alpha}}{1 - N_t}.$$

From firm labor demand:

$$W_t = (1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha}.$$

A positive Z_t or K_{t-1} shock shifts labor supply up by the same factor as labor demand:

$$Z_t K_{t-1}^\alpha.$$

Hence labor supply's wealth effect is exactly offset by labor demand increase. So when Z_t rises, the real wage increases. This generates a substitution effect toward more labor, but also an income effect toward more leisure.

$$W_t = (1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha}$$

$$Z_t \uparrow \Rightarrow MPN_t \uparrow \Rightarrow W_t \uparrow \Rightarrow \text{labor demand curve shifts up}$$

$$Z_t \uparrow \Rightarrow Y_t \uparrow \Rightarrow C_t = (1 - \alpha\beta) Y_t \uparrow \Rightarrow MU_{C,t} \downarrow \Rightarrow N_t \downarrow \quad \text{Wealth effect}$$

$$\begin{aligned} \underbrace{\frac{1 - \theta}{1 - N_t}}_{\text{marginal disutility of working}} &= \underbrace{\lambda_t W_t}_{\text{marginal utility value of wage income}} = \underbrace{MU_{C,t} W_t}_{\text{intra-temporal optimality}} = \frac{\theta}{C_t} (1 - \alpha) \frac{Y_t}{N_t} \\ &= \frac{\theta(1 - \alpha)}{N_t} \left(\frac{Y_t}{C_t} \right) = \frac{\theta(1 - \alpha)}{1 - \alpha\beta} \frac{1}{N_t}. \end{aligned}$$

Hansen (1985): Basic idea Expand log preference to make labor respond to shocks.

Household's problem:

$$\max_{\{C_t, L_t, N_t\}} E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, L_t) = E_0 \sum_{t=0}^{\infty} \beta^t u(C_t, 1 - N_t)$$

$$\text{s.t.} \quad N_t + L_t = 1$$

$$C_t + K_t = W_t N_t + R_t K_{t-1} + (1 - \delta) K_{t-1} \quad (\text{budget constraint not resource constraint})$$

$$K_t - (1 - \delta) K_{t-1} = I_t \quad \text{investment}$$

$$\mathcal{L} = E_0 \sum_{t=0}^{\infty} \beta^t \left[u(C_t, 1 - N_t) + \lambda_t (W_t N_t + R_t K_{t-1} + (1 - \delta) K_{t-1} - C_t - K_t) \right]$$

$$\frac{\partial \mathcal{L}}{\partial C_t} = 0 \quad \Rightarrow \quad \underbrace{U_{C,t}}_{\text{marginal utility of consumption}} = \lambda_t$$

$$\frac{\partial \mathcal{L}}{\partial N_t} = 0 \quad \Rightarrow \quad U_{N,t} + \lambda_t W_t = 0$$

$$\Rightarrow \quad - \underbrace{U_{N,t}}_{\text{marginal disutility of working}} = \underbrace{\lambda_t W_t}_{\text{marginal utility value of one more unit of wage income}}$$

$$\frac{\partial \mathcal{L}}{\partial K_t} = 0 \quad \Rightarrow \quad -\lambda_t \beta^t + \beta^{t+1} E_t [\lambda_{t+1} R_{t+1} + \lambda_{t+1} (1 - \delta)] = 0$$

$$\Rightarrow \quad U_{C,t} = \beta E_t [U_{C,t+1} (R_{t+1} + 1 - \delta)]$$

Firm's problem

$$\max_{N_t, K_{t-1}} \Pi_t = Y_t - W_t N_t - R_t K_{t-1}$$

s.t.

$$Y_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha}, \quad \log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t.$$

FOCs:

$$\frac{\partial \Pi_t}{\partial K_{t-1}} = 0 \quad \Rightarrow \quad \alpha Z_t K_{t-1}^{\alpha-1} N_t^{1-\alpha} = R_t \quad \Rightarrow \quad \alpha \frac{Y_t}{K_{t-1}} = R_t$$

$$\frac{\partial \Pi_t}{\partial N_t} = 0 \quad \Rightarrow \quad (1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha} = W_t \quad \Rightarrow \quad (1 - \alpha) \frac{Y_t}{N_t} = W_t.$$

Let utility function follow

$$U(C_t, 1 - N_t) = \ln C_t + A \ln(1 - N_t).$$

Then

$$U_{C,t} = \frac{1}{C_t}, \quad U_{N,t} = -\frac{A}{1 - N_t}.$$

Hence equilibrium conditions are

$$\left\{ \begin{array}{l} 1 = \beta E_t \left[\frac{C_t}{C_{t+1}} (R_{t+1} + 1 - \delta) \right], \\ \frac{A}{1 - N_t} = \frac{1}{C_t} W_t, \\ W_t = (1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha}, \\ R_t = \alpha Z_t K_{t-1}^{\alpha-1} N_t^{1-\alpha}, \\ \Pi_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha} - W_t N_t - R_t K_{t-1} = 0, \\ C_t + K_t = W_t N_t + R_t K_{t-1} + (1 - \delta) K_{t-1}, \\ \log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t. \end{array} \right.$$

Do not substitute out R_t, W_t , to make a system of equations of (C_t, N_t, K_t, Z_t) .

Substituting yields

$$\left\{ \begin{array}{l} 1 = \beta E_t \left[\frac{C_t}{C_{t+1}} (\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta) \right], \\ AC_t = (1 - N_t)(1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha}, \\ K_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha} + (1 - \delta) K_{t-1} - C_t, \\ \log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t. \end{array} \right.$$

Step 1: Steady state $\Rightarrow$ Step 2: Log-linearize $\Rightarrow$ Step 3: Numerically solve

Steady state

At steady state:

$$K_t = K_{t-1} = \bar{K}, \quad C_t = C_{t+1} = \bar{C}, \quad N_t = N_{t+1} = \bar{N}.$$

$$\bar{K} = \bar{Z} \bar{K}^\alpha \bar{N}^{1-\alpha} + (1 - \delta) \bar{K} - \bar{C} \Rightarrow \bar{C} = \bar{Z} \bar{K}^\alpha \bar{N}^{1-\alpha} - \delta \bar{K}. \quad (1)$$

$$1 = \alpha \beta \bar{Z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} + (1 - \delta) \beta. \quad (2)$$

$$A \bar{C} = (1 - \bar{N})(1 - \alpha) \bar{Z} \bar{K}^\alpha \bar{N}^{-\alpha}. \quad (3)$$

$$\Rightarrow \bar{Z} \bar{K}^\alpha \bar{N}^{1-\alpha} - \delta \bar{K} = \frac{1}{A} \left((1 - \alpha) \bar{Z} \bar{K}^\alpha \bar{N}^{-\alpha} - (1 - \alpha) \bar{Z} \bar{K}^\alpha \bar{N}^{1-\alpha} \right)$$

$$\Rightarrow \bar{Z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} + \frac{1 - \alpha}{A} \bar{Z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} = \frac{1 - \alpha}{A} \bar{Z} \bar{K}^{\alpha-1} \bar{N}^{-\alpha} + \delta$$

$$\Rightarrow \bar{Z} \bar{K}^{\alpha-1} \bar{N}^{1-\alpha} = \frac{A}{1 - \alpha + A} \left[\frac{1 - \alpha}{A} \bar{Z} \bar{K}^{\alpha-1} \bar{N}^{-\alpha} + \delta \right]$$

Using (2), after simplification one gets

$$\bar{N} = \left\{ 1 + \frac{A}{1 - \alpha} \left[1 - \frac{\alpha \beta \delta}{1 - \beta(1 - \delta)} \right] \right\}^{-1}.$$

Then

$$\bar{K} = \left[\frac{1 - \beta(1 - \delta)}{\bar{Z} \alpha \beta} \bar{N}^{\alpha-1} \right]^{\frac{1}{\alpha-1}} = \bar{N} \left[\frac{\bar{Z} \alpha}{\beta^{-1} - (1 - \delta)} \right]^{\frac{1}{1-\alpha}}.$$

And

$$\begin{aligned} \bar{C} &= \bar{Z} \bar{K}^\alpha \bar{N}^{1-\alpha} - \delta \bar{K} = \bar{N}^\alpha \cdot \left[\frac{\bar{Z} \alpha}{\beta^{-1} - (1 - \delta)} \right]^{\frac{\alpha}{1-\alpha}} \bar{N}^{1-\alpha} \bar{Z} - \delta \bar{K} \\ &= \bar{Z}^{\frac{1}{1-\alpha}} \bar{N} \left[\frac{\alpha}{\beta^{-1} - (1 - \delta)} \right]^{\frac{\alpha}{1-\alpha}} - \delta \bar{N} \bar{Z}^{\frac{1}{1-\alpha}} \left[\frac{\alpha}{\beta^{-1} - (1 - \delta)} \right]^{\frac{1}{1-\alpha}} \\ &= \bar{N} \bar{Z}^{\frac{1}{1-\alpha}} \left[\left(\frac{\alpha}{\beta^{-1} - (1 - \delta)} \right)^{\frac{\alpha}{1-\alpha}} - \delta \left(\frac{\alpha}{\beta^{-1} - (1 - \delta)} \right)^{\frac{1}{1-\alpha}} \right] \end{aligned}$$

Derive the log-linearized system

(i) TFP process:

$$\log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t$$

Let

$$Z_t = \bar{Z} e^{\hat{z}_t}.$$

Then

$$\begin{aligned} \hat{z}_t + \log \bar{Z} &= (1 - \psi) \log \bar{Z} + \psi \log \bar{Z} + \psi \hat{z}_{t-1} + \varepsilon_t \\ \Rightarrow \hat{z}_t &= \psi \hat{z}_{t-1} + \varepsilon_t. \end{aligned}$$

(ii)

$$K_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha} + (1 - \delta) K_{t-1} - C_t.$$

Substitute

$$K_t = \bar{K}(1 + \hat{k}_t), \quad Z_t = \bar{Z}(1 + \hat{z}_t), \quad K_{t-1} = \bar{K}(1 + \hat{k}_{t-1}), \quad N_t = \bar{N}(1 + \hat{n}_t), \quad C_t = \bar{C}(1 + \hat{c}_t),$$

and keep only first-order terms:

$$\bar{K} + \bar{K} \hat{k}_t = \bar{Z} \bar{K}^\alpha \bar{N}^{1-\alpha} (1 + \hat{z}_t) (1 + \alpha \hat{k}_{t-1}) (1 + (1 - \alpha) \hat{n}_t) + (1 - \delta) \bar{K} + (1 - \delta) \bar{K} \hat{k}_{t-1} - \bar{C} - \bar{C} \hat{c}_t.$$

Using the steady state condition and dropping higher-order terms:

$$\bar{K} \hat{k}_t = \alpha \bar{Y} \hat{k}_{t-1} + \bar{Y} \hat{z}_t + (1 - \alpha) \bar{Y} \hat{n}_t + (1 - \delta) \bar{K} \hat{k}_{t-1} - \bar{C} \hat{c}_t,$$

where

$$\bar{Y} \equiv \bar{Z} \bar{K}^\alpha \bar{N}^{1-\alpha}.$$

Hence

$$\hat{k}_t = \left(\alpha \frac{\bar{Y}}{\bar{K}} + 1 - \delta \right) \hat{k}_{t-1} + (1 - \alpha) \frac{\bar{Y}}{\bar{K}} \hat{n}_t + \frac{\bar{Y}}{\bar{K}} \hat{z}_t - \frac{\bar{C}}{\bar{K}} \hat{c}_t.$$

(iii)

$$AC_t = (1 - N_t)(1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha}.$$

Log-linearizing gives

$$\hat{n}_t = \left(\alpha + \frac{\bar{N}}{1 - \bar{N}} \right)^{-1} (\hat{z}_t + \alpha \hat{k}_{t-1} - \hat{c}_t).$$

(iv) Euler equation:

$$1 = \beta E_t \left[\frac{C_t}{C_{t+1}} (\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta) \right].$$

Let

$$M_{t+1} \equiv \alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha}, \quad G_{t+1} \equiv M_{t+1} + 1 - \delta.$$

Then

$$\hat{m}_{t+1} = \hat{z}_{t+1} + (\alpha - 1)\hat{k}_t + (1 - \alpha)\hat{n}_{t+1}.$$

Define

$$\tilde{\delta} \equiv \frac{\bar{M}}{\bar{G}}.$$

Since

$$\bar{G} = \beta^{-1}, \quad \bar{G} = \bar{M} + (1 - \delta),$$

we have

$$\tilde{\delta} = \frac{\bar{M}}{\bar{G}} = 1 - \frac{1 - \delta}{\bar{G}} = 1 - \beta(1 - \delta).$$

Thus

$$G_{t+1} \approx \bar{G}(1 + \tilde{\delta}\hat{m}_{t+1}).$$

Also

$$\frac{C_t}{C_{t+1}} = \frac{\bar{C}e^{\hat{c}_t}}{\bar{C}e^{\hat{c}_{t+1}}} = e^{\hat{c}_t - \hat{c}_{t+1}} \approx 1 + \hat{c}_t - \hat{c}_{t+1}.$$

So Euler equation becomes

$$1 = \beta E_t \left[\bar{G}(1 + \hat{c}_t - \hat{c}_{t+1} + \tilde{\delta}\hat{m}_{t+1}) \right]$$

Since

$$\beta\bar{G} = 1,$$

we obtain

$$E_t [\hat{c}_t - \hat{c}_{t+1} + \tilde{\delta}\hat{m}_{t+1}] = 0.$$

Hence

$$\hat{c}_t = E_t \hat{c}_{t+1} - \tilde{\delta} E_t [\hat{z}_{t+1} - (1 - \alpha)\hat{k}_t + (1 - \alpha)\hat{n}_{t+1}].$$

If you have CRRA,

$$\frac{C_t}{C_{t+1}} \rightarrow \left(\frac{C_t}{C_{t+1}} \right)^\sigma \Rightarrow \sigma \hat{c}_t = \sigma E_t \hat{c}_{t+1} - E_t \hat{r}_{t+1}.$$

Thus we have a system of equations:

$$\begin{cases} \hat{z}_t = \psi \hat{z}_{t-1} + \varepsilon_t, \\ \hat{c}_t = E_t \hat{c}_{t+1} - \tilde{\delta} E_t [\hat{z}_{t+1} - (1 - \alpha)\hat{k}_t + (1 - \alpha)\hat{n}_{t+1}], \\ \hat{n}_t = \left(\alpha + \frac{\bar{N}}{1 - \bar{N}} \right)^{-1} (\hat{z}_t + \alpha \hat{k}_{t-1} - \hat{c}_t), \\ \hat{k}_t = \left(\alpha \frac{\bar{Y}}{\bar{K}} + 1 - \delta \right) \hat{k}_{t-1} + (1 - \alpha) \frac{\bar{Y}}{\bar{K}} \hat{n}_t + \frac{\bar{Y}}{\bar{K}} \hat{z}_t - \frac{\bar{C}}{\bar{K}} \hat{c}_t. \end{cases}$$

Four equations, four variables:

$$(\hat{k}_{t-1}, \hat{c}_t, \hat{n}_t, \hat{z}_t).$$

$\hat{k}_{t-1}$ is predetermined variable, $\hat{z}_t$ follows a stable AR(1) process,

$(\hat{c}_t, \hat{n}_t)$ are jump variables.

Use undetermined coefficient method; guess

$$\hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t, \quad \hat{c}_t = V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t, \quad \hat{n}_t = V_{nk}\hat{k}_{t-1} + V_{nz}\hat{z}_t.$$

Then

$$\hat{k}_t = V_{kz} \sum_{i=0}^{\infty} \sum_{j=0}^i V_{kk}^j \psi^{i-j} \varepsilon_{t-i}, \quad \hat{z}_t = \sum_{s=0}^{\infty} \psi^s \varepsilon_{t-s},$$

and since

$$\hat{y}_t = \alpha \hat{k}_{t-1} + (1 - \alpha) \hat{n}_t + \hat{z}_t,$$

we can write

$$\hat{y}_t = V_{yz} \varepsilon_t + \sum_{i=0}^{\infty} \left(V_{yk} V_{kz} \sum_{j=0}^i V_{kk}^j \psi^{i-j} + V_{yz} \psi^{i+1} \right) \varepsilon_{t-i-1}.$$

Therefore,

$$\text{Var}(\hat{y}_t) = \left(V_{yz}^2 + \sum_{i=0}^{\infty} \left[V_{yk} V_{kz} \sum_{j=0}^i V_{kk}^j \psi^{i-j} + V_{yz} \psi^{i+1} \right]^2 \right) \text{Var}(\varepsilon_t).$$

Given parameters' value, we can back out the explicit numerical solution.

This model works:

- 1) well for consumption volatility
- 2) bad for output volatility: requires too small TFP volatility
- 3) bad for hours volatility $\Rightarrow$ way too low.
- 4) bad for investment volatility $\Rightarrow$ far too low.

But it allows N_t to move with shocks compared to Long-Plosser. While with this set of preference and technology, it cannot generate enough labor and investment volatility relative to output.

Hansen (1985) with indivisible labor

Labor market phenomena:

- 1) Hours fluctuate a lot relative to productivity.
- 2) Micro evidence does not support huge intertemporal substitution of labor at individual level.

Target: get big aggregate hours fluctuations without assuming each person is very willing to shift leisure across time.

Key mechanism:

- 1) Each individual's labor is a corner-choice:

$$\tilde{n}_t \in \{0, n^*\}.$$

Either work a fixed shift or no work.

- 2) Lottery over jobs: assigned a job with probability π_t .
- 3) Complete unemployment insurance: C_t is independent of job status.

Crucial: we can write expected utility clearly at the “representative agent” level.

Thus total labor supply:

$$H_t = \bar{L} \pi_t n^*$$

where $\bar{L}$ is total labor force.

Average labor supply:

$$n_t = \frac{H_t}{\bar{L}} = \pi_t n^*.$$

Representative-agent utility: individual utility is

$$U(C_t) + v(\tilde{n}_t).$$

Given lottery,

$$E[U(C_t) + v(\tilde{n}_t)] = U(C_t) + \pi_t v(n^*) + (1 - \pi_t)v(0).$$

Let

$$v(0) = 0, \quad \frac{v(n^*)}{n^*} = -A < 0.$$

Then

$$E[U(C_t) + v(\tilde{n}_t)] = U(C_t) + \pi_t(-An^*) = U(C_t) - An_t.$$

Suppose

$$U(C_t) = \ln C_t \quad \Rightarrow \quad U_t = \ln C_t - An_t.$$

Hence

$$U_{C,t} = \frac{1}{C_t}, \quad U_{N,t} = -A, \quad -U_{N,t} = U_{C,t}W_t$$

Micro-level: no smoothing in labor, either n^* or 0, keep small intertemporal labor substitution.

Macro-level: planner / representative agent can vary π_t a lot, aggregate hours become very responsive.

Thus, we can define the equilibrium conditions for this model:

$$\left\{ \begin{array}{l} 1 = \beta E_t \left[\frac{C_t}{C_{t+1}} (\alpha Z_{t+1} K_t^{\alpha-1} N_{t+1}^{1-\alpha} + 1 - \delta) \right], \\ K_t = Z_t K_{t-1}^\alpha N_t^{1-\alpha} + (1 - \delta) K_{t-1} - C_t, \\ AC_t = (1 - \alpha) Z_t K_{t-1}^\alpha N_t^{-\alpha}, \quad \text{key difference} \quad - U_{N,t} = U_{C,t}W_t = U_{C,t}MPL_t, \\ \log Z_t = (1 - \psi) \log \bar{Z} + \psi \log Z_{t-1} + \varepsilon_t. \end{array} \right.$$

Mechanism

In the indivisible labor model, aggregate hours fluctuate mainly through the employment margin: more people work rather than each individual continuously adjusting hours.

By contrast, in the divisible labor model,

$$TFP \uparrow \Rightarrow \text{demand for labor} \uparrow \Rightarrow W_t \uparrow \Rightarrow N_t \uparrow \Rightarrow 1 - N_t \downarrow$$

$\ln(1 - N_t)$ becomes steeper $\Rightarrow$ the cost of giving up a marginal hour of leisure rises

$\Rightarrow$ unwilling to adjust N_t , wage moves more $\Rightarrow$ stiff labor

$\Rightarrow$ small hour volatility. (each person chooses hours continuously)

Consistency with facts

$$n_t = \frac{H_t}{\bar{L}} = \frac{H_t}{L_t} \cdot \frac{L_t}{\bar{L}} \approx n^*(1 - u_t) = n^*\pi_t,$$

where u_t is the unemployment rate.

$$\frac{H_t}{L_t} = \bar{n}_t \approx n^* \quad \text{is empirically stable.}$$

*Cyclical movement in total hours comes from (extensive margin) employment, not from hours per worker (intensive margin).

Other RBC models

1. CRRA utility (King and Rebelo, 1999):

$$U(C_t, L_t) = \ln C_t + A \frac{(1 - N_t)^{1-\chi} - 1}{1 - \chi}.$$

Investment volatility $\uparrow\uparrow$ (almost correct), but underpredicts persistence, over-predicts comovement; consumption & labor are less volatile in model than data.

change preference curvature and more volatile TFP

2. KPR utility (King, Plosser and Rebelo, 1988):

$$U(C_t, L_t) = \frac{[C_t v(L_t)]^{1-\sigma} - 1}{1 - \sigma}.$$

Non-separable preference between C and N . $U_{C,t}$ depends on L_t , U_{L_t} depends on C_t . Relax the too-tight linkages $\Rightarrow$ improve comovement patterns.

3. GHH utility (Greenwood–Hercowitz–Huffman, 1988, AER)

$$U(C_t, N_t) = \frac{1}{1 - \sigma} \left[\left(C_t - A \frac{N_t^{1+\psi}}{1 + \psi} \right)^{1-\sigma} - 1 \right].$$

Cancel wealth effect, labor supply responds to substitution effect (wage/productivity), $\Rightarrow$ even more volatile N_t .

Since

$$-U_{N,t} = U_{C,t} W_t \quad \Rightarrow \quad A N_t^\psi = W_t.$$

4. Capital utilization:

$$Y_t = Z_t (u_t K_{t-1})^\alpha N_t^{1-\alpha},$$

$$K_t = I_t + [1 - \delta(u_t)] K_{t-1}, \quad \delta(u) > 0$$

Extra intensive margin for capital $\Rightarrow$ higher investment volatility, higher output volatility.